

CHALLENGES FACED AND SOLUTIONS IMPLEMENTED

Introduction

During the Raritone Summer Internship 2026, several technical and project-related challenges were encountered while working on backend validation, authentication support, security verification, API testing, workflow integration, documentation, and future technology planning. These challenges provided valuable learning opportunities and enhanced my problem-solving abilities. Through research, collaboration, testing, and systematic analysis, effective solutions were identified and implemented to ensure the successful development and validation of the backend system.

1. Understanding a Large Backend Architecture

Challenge

The Raritone Backend System consisted of multiple interconnected modules, APIs, middleware components, database models, and authentication workflows. Understanding how all components interacted with one another was initially challenging.

Solution

I studied the project structure, API flows, database relationships, and module responsibilities to gain a clear understanding of the overall architecture. Reviewing documentation and analyzing request-response workflows helped in understanding how the backend operated as a unified system.

Benefits

- Improved understanding of enterprise backend architecture
 - Better workflow analysis skills
 - Enhanced system-level thinking
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2. Validating Wishlist Module Workflows

Challenge

The Wishlist Module required verification of product addition, removal, retrieval, and user-product relationships while maintaining data consistency.

Solution

Comprehensive API testing and workflow validation were performed to verify that wishlist operations functioned correctly and maintained accurate relationships between users and saved products.

Benefits

- Improved module reliability
 - Enhanced user experience
 - Accurate wishlist management
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3. Supporting Virtual Try-On Workflow Validation

Challenge

The Virtual Try-On Module was designed as an AI-ready component, making it important to verify workflow structures and future integration readiness.

Solution

Request flows, API behavior, and workflow logic were reviewed and validated to ensure compatibility with future AI-powered virtual try-on services.

Benefits

- Improved AI readiness
 - Better workflow consistency
 - Future integration support
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4. Authentication Workflow Verification

Challenge

Authentication systems involve multiple components such as login processes, token generation, access control, and protected routes. Verifying that these mechanisms functioned correctly was essential.

Solution

JWT authentication workflows, protected APIs, and token validation mechanisms were tested to ensure secure access management across the platform.

Benefits

- Improved authentication reliability
 - Enhanced user security
 - Secure access management
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5. Security Validation

Challenge

Modern applications must protect user data and backend resources from unauthorized access and invalid requests.

Solution

Security mechanisms such as authentication checks, access control validation, protected routes, and request validation processes were tested and verified.

Benefits

- Strengthened backend security
 - Improved data protection
 - Reduced security risks
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6. API Testing Across Multiple Modules

Challenge

Testing numerous APIs across different modules while ensuring consistent functionality and response behavior was time-consuming.

Solution

Postman was extensively used to test endpoints, validate responses, verify CRUD operations, and ensure proper error handling across various modules.

Benefits

- Improved API reliability
 - Better system stability
 - Reduced functional defects
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7. Workflow Integration Verification

Challenge

Backend modules needed to communicate seamlessly with one another. Identifying inconsistencies between workflows required careful validation.

Solution

Workflow testing and integration verification were performed to ensure smooth communication between APIs, controllers, database models, and related services.

Benefits

- Improved module integration
 - Consistent system behavior
 - Enhanced application reliability
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8. Documentation Management

Challenge

As the project expanded, maintaining accurate documentation for APIs, workflows, testing activities, and implementation details became increasingly important.

Solution

Documentation was continuously reviewed and updated to ensure that project information remained accurate, organized, and useful for future development activities.

Benefits

- Better maintainability
- Improved knowledge sharing

- Easier onboarding for future developers
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9. AI Integration Planning

Challenge

The platform was designed to support future AI-powered features such as virtual try-on systems, avatar generation, and recommendation engines. Planning for these technologies without immediate implementation required careful architectural consideration.

Solution

Research was conducted on AI workflows, recommendation systems, avatar generation concepts, and virtual try-on architectures to understand future integration requirements.

Benefits

- Improved AI readiness
 - Better future planning
 - Enhanced innovation potential
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10. Scalability Research

Challenge

The platform must be capable of supporting future growth in users, products, media assets, and AI processing requests.

Solution

Research was conducted on technologies including Redis caching, queue-based processing, Docker containerization, Kubernetes orchestration, and cloud infrastructure scaling.

Benefits

- Better scalability planning
 - Improved performance awareness
 - Stronger future architecture preparation
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11. Adapting Academic Knowledge to Real-World Development

Challenge

Applying theoretical concepts learned in academics to a real-world backend project required a deeper understanding of practical software engineering practices.

Solution

Active participation in testing, validation, workflow analysis, security verification, and research activities helped bridge the gap between theoretical learning and practical implementation.

Benefits

- Improved practical development skills
 - Better problem-solving capabilities
 - Enhanced industry readiness
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Conclusion

Throughout the internship, various technical and organizational challenges were encountered while working on backend validation, authentication support, security verification, API testing, workflow analysis, documentation, AI planning, and scalability research. By applying systematic problem-solving approaches, conducting extensive testing, collaborating with team members, and continuously learning new concepts, these challenges were successfully addressed.

The experience not only improved my technical capabilities but also strengthened my analytical thinking, troubleshooting skills, documentation practices, and understanding of real-world software development. The lessons learned from overcoming these challenges will serve as a strong foundation for future academic projects, internships, and professional software engineering opportunities.